

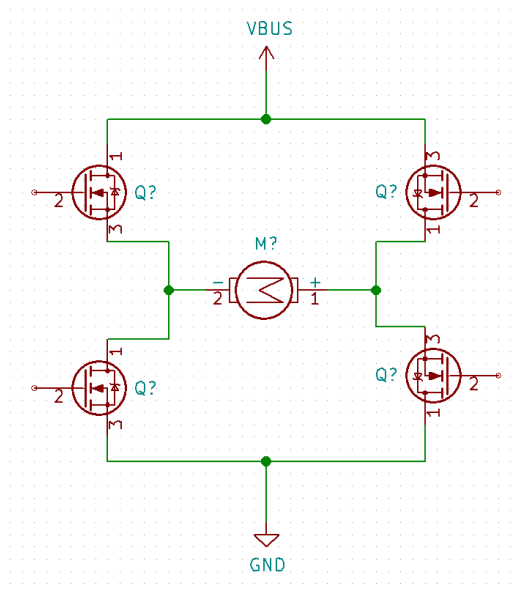
Modelling & Simulation: Assignment 4 (option 4)

H-bridge and PWM in motor drivers

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Introduction

In assignment 2, you modelled a motor controller, but simplified the motor driver to an **MSe**-element, as though it were a linear amplifier. In reality, this component is an H-bridge with high-power MOSFETs that use pulse width (or duty cycle) modulation to control the amount of power that goes to the motor.



The question is whether a real H-bridge can indeed be represented by a modulated voltage source. The goal of this assignment is to model an H-bridge, using modulated resistors as (only slightly) simplified models for the FETs, and comparing its behaviour with the simplified model of an **MSe**-element.

Assignment details on second page.

Assignment

Model an H-bridge for a (brushed) DC motor, find a way to implement PWM, and compare its behaviour with a linear amplifier, i.e. a modulated voltage source.

You should think of some simulation experiments through which you can find out the difference between the two models.

Some things to consider

1. Model your H-bridge after a real motor controller, for example a “Maxon ESCON Module 50/5.” This should give some ideas about resistance and PWM frequency.
2. When the PWM signal is “off,” are the FETs opened or is the motor “short-circuited” to keep the current flowing?
3. Sometimes, motor drivers include inductors that are placed in series with the motor. (See for example the ESCON 50/5, which has three 20 μH coils) Why is this? Can you show the effect in your simulation?
4. What is the behaviour of the H-bridge at very low (1 %) or very high (99 %) “power?”

Additional questions you could try to answer are:

- Do the energy losses in the system depend on the PWM frequency?
- Is the energy loss dependent on the inductance of the motor (plus optional series inductors)?
- If you include a closed-loop current controller as in assignment 2, do you notice any difference in the behaviour? (Do you reach the same “performance” with the same controller parameters, or with the re-tuned current controller gains?)